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The Skill Complementarity of Broadband Internet

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1 Big picture

- **Question.** Does broadband internet adoption in firms raise productivity and wages, and if so is the technological change skill-biased or factor-neutral?
- **Setting.** The paper studies Norway from 2001 to 2007 and exploits a public broadband rollout that created staggered variation in local broadband availability across municipalities.
- **Core challenge.** Firms choose whether to adopt broadband, and places that get broadband earlier might also have different underlying trends in productivity, wages, or labor demand. So a simple correlation between broadband and outcomes is not causal.
- **Main idea.** First estimate intention-to-treat effects of *availability* of broadband in a municipality. Then use household availability as an instrument for firm-level broadband adoption to estimate how adoption shifts the production function.
- **Main worker-side result.** Skilled workers benefit, while less skilled workers do not. A 10 percentage point increase in broadband availability raises skilled workers' hourly wages by about 0.18% and their employment by about 0.20 percentage points in the two-skill specification. In the three-skill specification, low-skilled wages fall and skilled wages rise.
- **Main firm-side result.** Broadband availability increases the output elasticity of skilled labor and reduces the output elasticity of low-skilled labor. In the baseline two-skill specification, the interaction of availability with skilled labor is positive, while the interaction with unskilled labor is negative and much smaller.
- **Main IV production-function result.** Broadband adoption in firms is a skill-biased technological change. In the IV estimates, adoption raises the output elasticity on skilled labor and lowers it for unskilled or low-skilled labor, while capital does not move much.
- **Mechanism.** Broadband appears to complement nonroutine abstract tasks and substitute for routine tasks. That is the paper's preferred task-based interpretation of the skill bias.
- **Why this matters.** The paper is not just about internet infrastructure. It is about how a specific ICT shock changes the demand for skills, the organization of production, and the distribution of rents between firms and workers.
- **Economic takeaway.** Broadband adoption seems to raise firm productivity, but the gains are not evenly shared. Skilled labor and firms gain; low-skilled labor loses. This distributional pattern can itself become a political barrier to broadband expansion.

2 Data and measurement (key objects)

2.1 Main data sources

- **Firm data.** Administrative registers for all nonfinancial joint-stock firms from 2000 to 2008. These contain revenues, value added, intermediates, capital, wage bills, industry codes, and municipality identifiers.
- **Linked employer-employee data.** The authors observe each worker's education and annual labor income, which allows them to split labor into skill groups and connect workers to firms and municipalities.

- **Broadband adoption data.** For 2001 to 2007, they observe whether firms subscribe to broadband in the annual ICT Usage of Firms survey, a stratified random sample of firms by industry and employment size.
- **Broadband availability data.** Municipality-level availability rates of broadband to households. This is the key policy-driven source of variation and the instrument for firm adoption.
- **Hourly wage and occupation data.** Statistics Norway’s wage statistics survey provides hourly wages and occupation codes. The occupation codes are later linked to task intensities from the DOT data set.

2.2 Sample definition and units of observation

- **Worker regressions.** The employment sample is the full population aged 18 to 67. The wage sample is workers aged 18 to 67 in the wage survey.
- **Firm regressions.** The intention-to-treat firm regressions use the population of joint-stock firms with at least one skilled and one unskilled worker. The adoption and IV production-function analysis uses the internet survey sample where broadband adoption is observed.
- **Population versus survey sample.** The population sample contains 149,676 firm-year observations. The survey sample for adoption and IV contains 16,744 firm-year observations.

2.3 How the paper defines skill

- **Baseline split.** Skilled means college or university degree; unskilled means less than college.
- **Refined split.** In some specifications, unskilled is split into medium skilled (high-school graduates without college) and low skilled (no high-school diploma).
- **Task interpretation.** Occupations are mapped into nonroutine abstract, routine, and nonroutine manual task intensities.

2.4 Key outcome variables

- Worker outcomes: log hourly wage and employment status.
- Firm outcome: log value added.
- Input measures in production functions: log capital and log labor inputs, where labor is measured by skill-specific wage bills.
- Adoption outcome: a broadband subscription dummy at the firm-year level.

2.5 Why wage bills are used as labor inputs

- The paper measures labor input with wage bills rather than headcounts.
- This is meant to put labor and capital on a more comparable basis: capital is measured in money terms, so labor is also measured in money terms to better capture quality differences across workers.

2.6 Descriptive statistics worth remembering

- **Employment rates.** In 2001, employment rates are 0.67 for unskilled and 0.85 for skilled workers; overall averages over the sample are 0.65 and 0.84.
- **Log hourly wages.** In 2001, unskilled workers average 2.86 and skilled workers 3.10; by 2007 they are at 3.01 and 3.27.
- **Firm size and inputs.** Mean log value added is about 6.88 overall, mean log capital about 4.49, mean log unskilled wage bill 5.63, and mean log skilled wage bill 4.19.
- **Broadband definition.** Broadband means download speeds above 256 kbit/s. Before rollout, firms mainly had dial-up access.

2.7 What is observed and what is not

- The authors do *not* observe availability of broadband to firms directly. They observe availability to households and use it as an instrument for firm adoption.
- They observe output and inputs, but productivity is partly unobserved to the econometrician. That is why Levinsohn-Petrin style control-function methods matter in the production-function estimation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline intention-to-treat regression

The core reduced-form specification is

$$y_{imt} = x'_{imt}\beta_0 + z_{mt}x'_{imt}\beta_1 + w'_{imt}\gamma + \mu_m + \tau_t + u_{imt}.$$

- z_{mt} is broadband availability in municipality m and year t .
- μ_m are municipality fixed effects; τ_t are year fixed effects.
- In worker regressions, x_{imt} is a vector of education dummies and w_{imt} contains gender, experience, experience squared, and industry controls.
- In firm regressions, x_{imt} contains log inputs such as capital and skill-specific labor.

Interpretation. This is the paper's main difference-in-differences style design with a continuous treatment. The coefficients in β_1 tell us whether broadband availability changes wages, employment, or output elasticities differently by skill group.

3.2 Timing-of-rollout specification

To check whether rollout timing is related to pre-existing municipality characteristics, the paper estimates a specification of the form

$$\Delta z_{mt} = \alpha_m + (\tau_t \times w_m)' \phi + e_{mt},$$

where w_m includes baseline municipality covariates such as income, education, urbanization, unemployment, industrial composition, skill composition, and pre-rollout growth in wages and employment.

Interpretation. This is not the main estimating equation. It is a diagnostic regression asking whether municipalities with different baseline characteristics systematically got broadband earlier.

3.3 Event-study style illustration

The paper recenters time so that period 0 is the year with the largest increase in broadband availability in a municipality, then plots changes in output elasticities and the skill premium around that event.

Interpretation. This is meant to show the basic dynamics visually: when availability jumps, the output elasticity of skilled labor rises, that of unskilled labor falls, and the return to skill goes up.

3.4 Production function with endogenous technology adoption

To interpret broadband as technological change, the paper estimates

$$y_{imt} = x'_{imt}\delta_0 + D_{imt}x'_{imt}\delta_1 + w'_{imt}\gamma + \mu_m + \tau_t + \varepsilon_{imt},$$

where D_{imt} equals one if firm i has broadband in year t .

- The coefficients in δ_0 are the pre-adoption Cobb-Douglas elasticities.
- The coefficients in δ_1 tell us how adoption changes the intercept and the factor elasticities.

Interpretation. If broadband is skill-biased, then adoption should raise the elasticity on skilled labor, reduce the elasticity on unskilled labor, or both.

3.5 First-stage system for the IV production function

Because D_{imt} and the interacted terms $D_{imt} \times x_{k,imt}$ are endogenous, the paper instruments them with availability and availability interacted with covariates:

$$\begin{aligned} D_{imt} &= x'_{imt}\alpha + z_{mt}x'_{imt}\eta + w'_{imt}\rho + \mu_m + \tau_t + \nu_{imt}, \\ D_{imt}x_{k,imt} &= x'_{imt}\alpha_k + z_{mt}x'_{imt}\eta_k + w'_{imt}\rho_k + \mu_{km} + \tau_{kt} + \nu_{k,imt}. \end{aligned}$$

Interpretation. Availability to households is used as an instrument for firm adoption, and interacted instruments are used for the endogenous interaction terms in the production function.

3.6 Reduced-form adoption equation

The paper also estimates a simpler adoption regression:

$$D_{imt} = \alpha z_{mt} + w'_{imt}\rho + \mu_m + \tau_t + \nu_{imt}.$$

Interpretation. This tells us whether the public rollout actually shifted firm behavior. The estimated coefficient is about 0.23, meaning a 10 percentage point increase in availability raises firm broadband adoption by about 2.3 percentage points.

3.7 Task-based wage regressions

To interpret the skill bias more structurally, the paper estimates wage regressions with task intensities and their interaction with broadband availability:

$$\log w_{imt} = \theta_0 + \theta_A A_{imt} + \theta_R R_{imt} + \theta_M M_{imt} + \psi_A (A_{imt} \times z_{mt}) + \psi_R (R_{imt} \times z_{mt}) + \psi_M (M_{imt} \times z_{mt}) + \text{controls} + \mu_m + \tau_t + \xi_{imt}$$

Here A , R , and M denote abstract, routine, and manual task intensities.

Interpretation. If broadband complements abstract tasks and substitutes for routine tasks, then $\psi_A > 0$ and $\psi_R < 0$.

3.8 Levinsohn-Petrin production-function correction

The paper notes that factor inputs may be correlated with unobserved productivity. To address this, it follows Levinsohn and Petrin (2003), using intermediate inputs as a proxy for unobserved productivity in the production-function estimation.

Interpretation. The baseline intention-to-treat results are already informative, but the LP correction strengthens the case that the production-function results are not just driven by endogenous input choices.

3.9 How to read output elasticities economically

- A higher coefficient on skilled wage bills in a Cobb-Douglas log-value-added regression means skilled labor contributes more to output at the margin.
- But changes in marginal productivity are not identical to changes in wages, because broadband can also shift the intercept and other factor elasticities.
- The authors show that only about 20% of the increase in the marginal productivity of skilled workers is passed through to wages in the short run, suggesting substantial rents to firms.

4 Identification and instruments (what makes the estimates credible)

4.1 Core identification idea

- The key source of variation is the staggered rollout of broadband infrastructure across municipalities.
- Because rollout depended heavily on limited funding, geography, topography, and pre-existing infrastructure, it generated spatial and temporal variation in availability that the authors argue is plausibly exogenous once municipality and year fixed effects are included.

4.2 Why municipality and year fixed effects matter

- Municipality fixed effects absorb permanent local differences such as geography, baseline industrial structure, and long-run development.

- Year fixed effects absorb national macro shocks and common technological trends.
- Identification then comes from within-municipality changes over time relative to the national average rollout path.

4.3 Evidence that rollout timing is not driven by observables

- The authors report that 84% of the variation in availability is explained by time-invariant municipality characteristics and common time effects.
- Only about 1% is explained by a large set of time-varying local demand and supply variables.
- Rollout timing is not systematically related to baseline output, inputs, industrial structure, skill composition, or pre-rollout wage and employment growth; early on there is some relationship with urbanization, but it disappears after 2002.

4.4 Why the event-study graphs are supportive but not decisive

- Figure II shows sharp co-movement between jumps in availability and changes in output elasticities and the skill premium.
- But the authors are explicit that these graphs are only suggestive; they still rely on the regression evidence with controls and fixed effects.

4.5 Robustness to differential trends

The paper checks the baseline results in multiple ways:

- add rich time-varying demographic and industry controls,
- interact baseline municipality characteristics with linear and quadratic trends,
- include municipality-specific linear trends,
- exclude Oslo and the three largest cities,
- redefine local labor markets at the broader regional level,
- use pre-rollout inputs or Levinsohn-Petrin corrections in production-function estimation.

Bottom line. The sign pattern — skilled up, unskilled down — is stable.

4.6 Exclusion restriction for the IV analysis

The stronger IV interpretation requires that availability affects productivity and wages only through firm adoption, not through some direct household-side channel.

The paper challenges this restriction using several arguments:

- **Demand-side story.** If household broadband just shifted product demand, the effect should weaken in tradable sectors where demand is tied to world markets. It does not.

- **Online orders.** There is no significant effect of availability on the probability that firms receive orders online.
- **Broadband-sector firms.** Results are robust when telecom and IT-consulting firms are excluded.
- **General technical upgrading.** There is no evidence that broadband availability simply proxies for more PC use in firms.
- **Work from home.** A work-from-home story would not naturally explain the negative effect on low-skilled productivity.

4.7 Placebo test with always takers and never takers

Some firms had their own broadband networks before the public rollout, while some never adopted even after availability became universal.

- If availability had a direct effect unrelated to firm adoption, it should affect these firms too.
- Table VI shows no significant availability effects on value added elasticities in the always-/never-taker subsample.

Interpretation. This is one of the cleanest pieces of support for the exclusion restriction.

4.8 Who are the compliers?

- Firms with more skilled workers, more PC use, and larger value added are more likely to adopt broadband when it becomes locally available.
- So the IV is identified off a selective set of early adopters that already possess complementary organizational or skill inputs.

Interpretation. The IV estimates are local to firms for which the public rollout actually changed adoption. They are informative, but not automatically the long-run effect for every firm in the economy.

4.9 Main limitations

- The exact marginal-productivity effects from the IV production function are imprecisely estimated.
- Occupation-based task measures are only observed for about 52% of the wage-survey sample.
- The IV estimates are local to early adopters, which limits external extrapolation.
- The paper can show that broadband adoption shifts production technology, but the detailed within-firm organizational channel remains only partly observed.

5 Tables and figures

5.1 Table I and Table II: what the samples look like

	2001	2004	2007	Overall
Employment rate				
Total	0.72 (0.45)	0.70 (0.46)	0.72 (0.45)	0.71 (0.45)
Unskilled	0.67 (0.47)	0.64 (0.48)	0.66 (0.47)	0.65 (0.48)
Skilled	0.85 (0.35)	0.83 (0.37)	0.85 (0.36)	0.84 (0.37)
Number of individuals, aged 18–67	2,829,739	2,899,342	2,991,389	20,327,515
Log hourly wage (in USD)				
Total	2.95 (0.27)	3.05 (0.28)	3.12 (0.30)	3.04 (0.30)
Unskilled	2.86 (0.23)	2.94 (0.24)	3.01 (0.25)	2.93 (0.26)
Skilled	3.10 (0.27)	3.20 (0.28)	3.27 (0.29)	3.18 (0.30)
Number of workers in wage survey, aged 18–67	1,161,912	1,246,036	1,349,481	8,759,388

The employment rates are based on the entire population of individuals between the ages of 18 and 67. The hourly wages are based on the sample to workers aged 18–67 who are recorded in the wage statistics survey. (Un)Skilled comprises workers with(out) a college degree. Detailed descriptions of the variables are given in Online Appendix Table A.1.

	2001	2004	2007	Overall
Log inputs and output (USD, thousands)				
Revenues	7.63 (1.31)	7.63 (1.31)	7.74 (1.38)	7.65 (1.33)
Value added	6.83 (1.20)	6.86 (1.20)	6.97 (1.28)	6.88 (1.22)
Intermediates	6.61 (1.93)	6.59 (1.93)	6.68 (1.98)	6.61 (1.94)
Capital	4.68 (1.89)	4.45 (1.98)	4.48 (2.02)	4.49 (1.97)
Log wage bills (USD, thousands)				
Total	5.90 (1.15)	5.98 (1.16)	6.19 (1.20)	6.01 (1.17)
Unskilled	5.52 (1.26)	5.60 (1.25)	5.80 (1.29)	5.63 (1.26)
Skilled	4.08 (1.64)	4.17 (1.69)	4.36 (1.77)	4.19 (1.70)
Number of firms				
Population	19,598	21,441	23,282	149,676
Survey	2,118	2,270	3,093	16,744

This table shows summary statistics for the population of joint-stock firms over the years 2001–2007, consisting of all joint-stock firms with at least one unskilled and one skilled employee. (Un)Skilled comprises workers with(out) a college degree. Detailed descriptions of the variables are given in Online Appendix Table A.1.

Read: These tables establish the scale of the worker and firm data and the basic skill differences in the Norwegian labor market.

- Table I shows that skilled workers have both higher employment rates and higher hourly wages throughout the sample.
- Table II shows that firms in the production-function sample are sizable, and that skilled wage bills rise steadily over time.
- The survey sample used for adoption and IV is much smaller than the population sample but is weighted to be representative.

Why it matters. The later results are easier to interpret once one sees that the skilled-unskilled gap is already economically meaningful before broadband expansion.

5.2 Figure I: the rollout really was staggered geographically

Read: Figure I maps household broadband availability in 2001, 2003, and 2005.

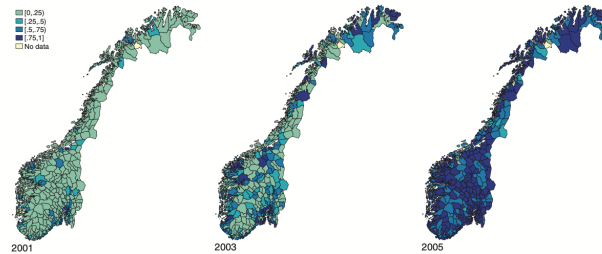


FIGURE I
Geographical Distribution of Broadband Availability Rates
The graphs show the geographical distribution of broadband availability rates of households in 2001, 2003, and 2005.

- In 2001, only a small set of municipalities around major cities had meaningful coverage.
- By 2005, availability is much broader, but the pattern remains far from a single national jump.
- Variation exists both across municipalities and within municipalities over time.

Why it matters. This figure is the visual foundation for the identification strategy. The paper needs staggered timing, not one common national adoption date.

5.3 Figure II: event-study illustration of skill complementarity

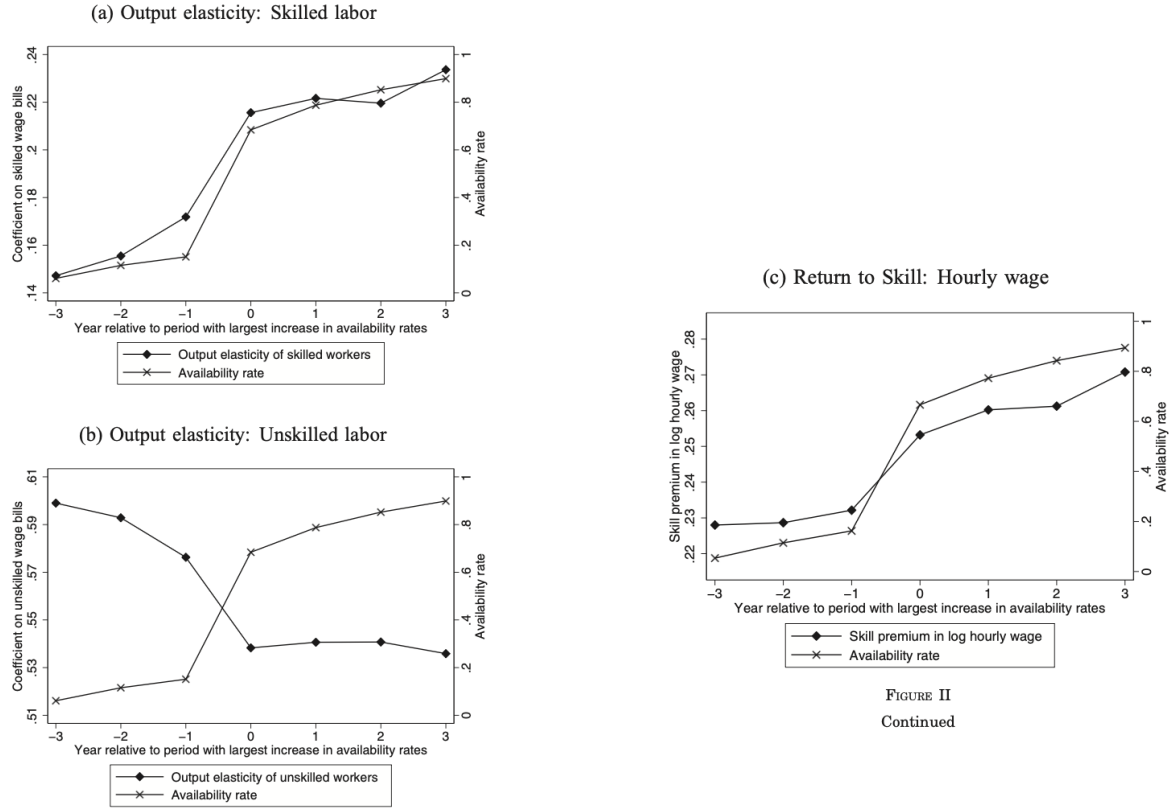


FIGURE II

Output Elasticities and Skill Premiums, Before and After the Largest Increase in Availability Rates (Period 0)

Period 0 represents the year with the strongest growth in availability rates in a given municipality. In each period, we estimate Cobb-Douglas production functions and wage regressions. Graphs (a) and (b) report period-specific OLS estimates of the output elasticity of skilled and unskilled labor. Graph (c) reports period-specific OLS estimates of log hourly wage on a dummy for skilled and controls for gender and potential experience.

Read: The figure centers municipalities at the year with the largest increase in availability.

- The output elasticity of skilled labor rises sharply around period 0, from roughly 0.15 before the jump to above 0.21 right after it, and ends around 0.23.
- The output elasticity of unskilled labor falls from around 0.60 before the jump to about 0.54 after it.
- The skill premium in hourly wages rises in parallel.

TABLE III
INTENTION-TO-TREAT EFFECTS ON WAGES AND EMPLOYMENT

Dependent variable	(1)	(2)	(3)	(4)
	Log hourly wage		Employment	
	2 skills	3 skills	2 skills	3 skills
Unskilled	2.939*** (0.00455)		0.691*** (0.00262)	
Low skilled		2.905*** (0.00431)		0.664*** (0.00231)
Medium skilled		2.977*** (0.00454)		0.731*** (0.00288)
Skilled	3.169*** (0.00420)	3.171*** (0.00407)	0.734*** (0.00480)	0.737*** (0.00477)
Availability × Unskilled	−0.00622 (0.00455)		0.000794 (0.00252)	
Low skilled		−0.0108*** (0.00325)		−0.00392 (0.00244)
Medium skilled		−0.00793 (0.00600)		0.00388 (0.00281)
Skilled	0.0178** (0.00720)	0.0202*** (0.00692)	0.0208** (0.00920)	0.0225** (0.00892)
Worker-year observations	8,759,388	8,759,388	20,327,515	20,327,515
		<i>p</i> -values		
Test for no skill bias	.000	.000	.012	.001

Notes. * $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on the model in equation (1), using worker-year observations over the period 2001–2007. Columns (1) and (2) consider the sample of workers aged 18–67 who are recorded in the wage statistics survey; the dependent variable is the log hourly wage in a given year. Columns (3) and (4) consider the entire population of individuals between the ages of 18 and 67; the dependent variable is an employment dummy, taking the value of 1 if the individual is employed in a given year. (Un)Skilled comprises workers with(out) a college degree. Low skilled comprises individuals without high school diploma, and medium skilled consists of high school graduates (without a college degree). All regressions include fixed effects for year, municipality, and industry and controls for gender, years of experience, and years of experience squared. The standard errors are clustered at the municipality level and robust to heteroskedasticity. We report p -values from two-sided tests of the null hypothesis that the coefficient on availability × log skilled is equal to the coefficient on availability × log unskilled (or availability × log low skilled).

Why it matters. It provides a simple picture of the central pattern: broadband expansion is associated with better relative outcomes for skilled labor.

5.4 Table III and Figure III: worker-side intention-to-treat effects

Read: Table III gives the main reduced-form worker results; Figure III converts them into actual-versus-counterfactual paths.

- In the two-skill wage regression, the availability interaction is 0.0178 for skilled workers and essentially zero for unskilled workers.
- In the three-skill wage regression, the interaction is −0.0108 for low skilled, −0.00793 for medium skilled, and 0.0202 for skilled workers.
- In employment, the skilled interaction is positive, about 0.0208 in the two-skill specification and 0.0225 in the three-skill specification.
- Unskilled employment barely moves.

Translate the magnitude.

- A 10 percentage point increase in availability raises skilled wages by about 0.18% to 0.20%.
- It lowers low-skilled wages by about 0.11%.
- It raises skilled employment by about 0.20 percentage points.

Figure III insight.

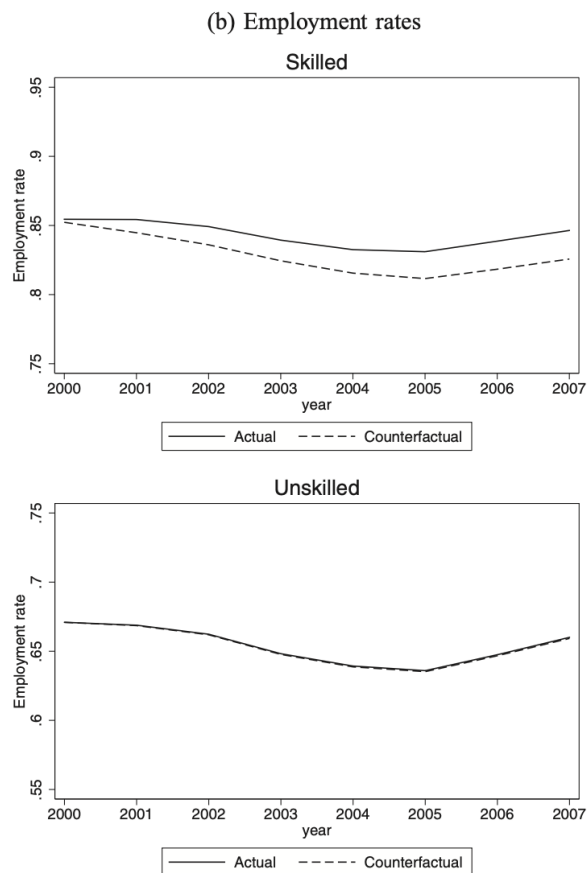
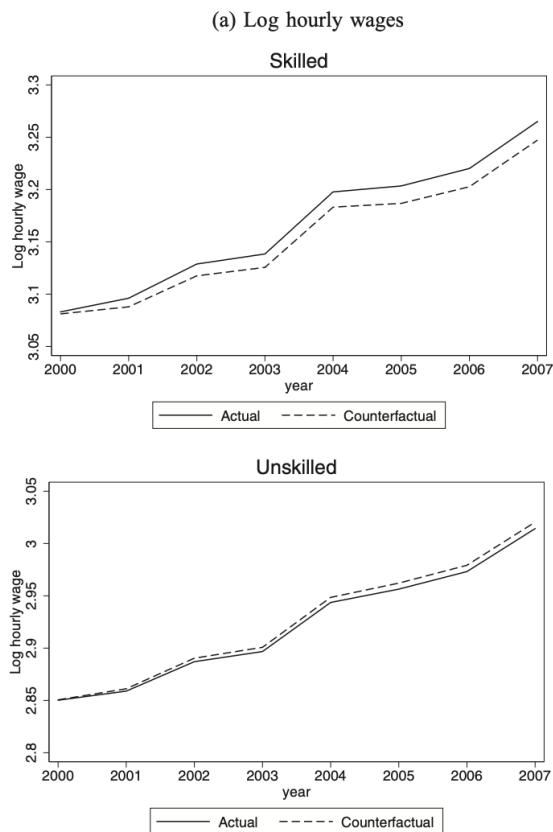


FIGURE III

Actual and Counterfactual Trends in Labor Market Outcomes

Solid line=actual outcome. Dashed line=counterfactual outcome in the absence of broadband internet expansion. A counterfactual outcome is measured as the actual outcome minus the predicted effect of broadband availability on the outcome. For comparability, in each graph the vertical axis covers four standard deviations of the labor market outcome across municipalities and years.

FIGURE III

Continued

TABLE IV
INTENTION-TO-TREAT EFFECTS ON OUTPUT ELASTICITIES

Dependent variable	(1)	(2)
	Log value added	
	2 skills	3 skills
Intercept	3.880*** (0.0965)	4.537*** (0.0791)
Log capital	0.100*** (0.00495)	0.0981*** (0.00490)
Log unskilled	0.576*** (0.0116)	
Log low skilled		0.298*** (0.00804)
Log medium skilled		0.265*** (0.00684)
Log skilled	0.136*** (0.00678)	0.134*** (0.00636)
Availability × Intercept	−0.500*** (0.111)	−0.561*** (0.0976)
Log capital	−0.00169 (0.00750)	0.000188 (0.00661)
Log unskilled	−0.0226 (0.0234)	
Log low skilled		−0.0274*** (0.00934)
Log medium skilled		0.0179* (0.00967)
Log skilled	0.0755*** (0.0166)	0.0645*** (0.0137)
Firm-year observations	149,676	137,498
Test for no skill bias	.012	<i>p</i> -values .000

Notes. * $p < .10$, ** $p < .05$, *** $p < .01$. Estimates are based on the model in equation (1), using the population of joint-stock firms over the period 2001–2007. The dependent variable is the log value added in a given year. (Un)Skilled comprises workers with(out) a college degree. Low skilled comprises individuals without high school diploma and medium skilled consists of high school graduates (without a college degree). All regressions include fixed effects for year, municipality, and industry. The standard errors are clustered at the municipality level and robust to heteroskedasticity. The hypothesis tests report p -values from two-sided tests of the null hypothesis that the coefficient on availability × log skilled is equal to the coefficient on availability × log unskilled (or availability × log low skilled).

- By 2007, skilled wages are about 1.8% higher than in the no-broadband counterfactual.
- Unskilled wages are about 0.6% lower.
- Skilled employment in 2005 is about 1.9 percentage points higher than the counterfactual.

Why it matters. These are not huge short-run effects year by year, but they cumulate and they all point in the same direction: broadband raises the relative demand for skill.

5.5 Table IV: firm-side intention-to-treat effects on output elasticities

Read: This table is the firm-side mirror image of Table III.

- In the two-skill specification, the availability interaction is −0.0226 on unskilled wage bills and 0.0755 on skilled wage bills.
- In the three-skill specification, the interaction is −0.0274 for low-skilled, 0.0179 for medium-skilled, and 0.0645 for skilled wage bills.
- The capital elasticity barely changes.

Interpretation. Broadband availability changes the slope of the production function in favor of skilled labor. It is not just that firms expand overall output; they also become more skill-intensive at the margin.

Extra point. When the authors translate elasticity shifts into implied marginal-productivity

changes, they conclude that only about 20% of the increase in skilled workers' marginal productivity is passed through to wages in the short run. That suggests firms retain a substantial share of the surplus.

5.6 Figure IV and Table V: first stage and who adopts first

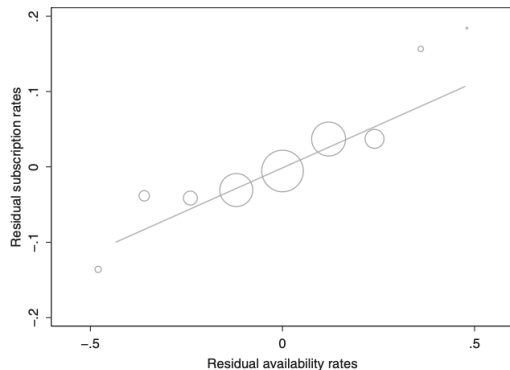


FIGURE IV

Association between Broadband Availability and Usage Rates, after Taking Out Municipality, Industry, and Year Fixed Effects

This figure uses the survey sample of joint-stock firms. Bins are based on the residual subscription rates of firms with bin size = 0.12. The size of the circle represents the number of firms in each bin. Sampling weights are used to ensure representative results for the population of joint-stock firms. The y-axis reports residuals from a regression of broadband subscription rates of firms on municipality, industry, and year fixed effects. The x-axis reports residuals from a regression of broadband availability rates of households on municipality, industry and year fixed effects.

TABLE V
CHARACTERIZING COMPLIER FIRMS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Composition of sample	$\hat{\delta}$	Composition of compliers	Log value added	Share of skilled workers	Share of workers using PC	Number of obs.
<i>Manufacturing</i>							
Low	0.19	0.17	0.16	14.16	0.11	0.32	2,216
High	0.19	0.20	0.19	14.35	0.31	0.58	2,218
<i>Wholesale</i>							
Low	0.22	0.21	0.23	13.51	0.11	0.57	2,609
High	0.22	0.24	0.26	13.69	0.32	0.73	2,609
<i>Services</i>							
Low	0.09	0.15	0.07	13.51	0.26	0.65	1,079
High	0.09	0.18	0.08	13.87	0.70	0.96	1,080
<i>Overall</i>	1.00	0.23	1.00	13.75	0.25	0.54	16,744

Notes. We partition the survey sample of joint-stock firms into six mutually exclusive groups by industry (three largest industries) and skill intensity (above and below median within each industry). Column (1) displays the proportion of the sample in each industry-skill intensity group. Column (2) reports estimates of $\hat{\delta}$ from equation (4) for each group. The proportion of the compliers of a given type is then calculated as the ratio of $\hat{\delta}$ for that subgroup to the $\hat{\delta}$ in the overall sample, multiplied by the proportion of the sample in the industry-skill group. Column (3) shows the distribution of the compliers by industry and skill intensity. Columns (4)–(6) report baseline characteristics of each industry-skill group. Sampling weights are used to ensure representative results for the population of joint-stock firms.

Read: Figure IV plots firm subscription residuals against household availability residuals after removing municipality, industry, and year fixed effects. Table V characterizes the firms whose adoption is most sensitive to availability.

- The first stage is strong and economically meaningful.
- The adoption coefficient is about 0.23, so a 10 percentage point increase in availability raises firm adoption by roughly 2.3 percentage points.
- Complier firms are disproportionately high-skill, more computer-intensive, and larger in value added.

Why it matters. Broadband is not adopted randomly. It is adopted faster where complementary organizational inputs already exist. This helps explain why the IV estimates point to a strong skill-biased effect.

5.7 Table VI: placebo test using always takers and never takers

Read: The authors re-estimate the reduced-form productivity regression on firms whose adoption status could not have been changed by local availability.

- In the baseline sample, availability interacts positively with skilled labor and negatively with the intercept.
- In the always-/never-taker sample, those interaction effects disappear.

Why it matters. If availability had direct effects unrelated to adoption, these firms should respond too. They do not. This gives extra credibility to the exclusion restriction.

TABLE VI
PLACEBO TEST: OUTPUT ELASTICITIES

Dependent variable	Log value added	
	Baseline sample of firms	Always/never taker firms only
Intercept	3.880*** (0.0965)	4.388*** (0.692)
Log capital	0.100*** (0.00495)	0.114*** (0.0313)
Log unskilled	0.576*** (0.0116)	0.505*** (0.0869)
Log skilled	0.136*** (0.00678)	0.171*** (0.0295)
Availability × Intercept	−0.500*** (0.111)	−0.212 (0.709)
Log capital	−0.00169 (0.00750)	−0.0230 (0.0345)
Log unskilled	−0.0226 (0.0234)	0.0295 (0.0860)
Log skilled	0.0755*** (0.0166)	0.00944 (0.0278)
Firm-year observations	149,676	2,233

Notes. Estimates are based on the model in equation (1). Column (1) considers the population of joint-stock firms, while column (2) restricts the sample to firms that have adopted broadband even when the household availability rate is 0 (always takers) and firms that have not adopted broadband even when the household availability rate is 1 (never takers). (Un)Skilled comprises workers with(out) a college degree. All regressions include fixed effects for year, municipality and industry. The standard errors are clustered at the municipality level and robust to heteroskedasticity.

TABLE VII
BROADBAND ADOPTION AND TECHNOLOGICAL CHANGE

Dependent variable	(1)	(2)	(3)	(4)
	Log value added			
	2 skills		3 skills	
	OLS	IV	OLS	IV
Intercept	3.809*** (0.145)	3.751*** (0.455)	4.695*** (0.120)	4.636*** (0.437)
Log capital	0.0987*** (0.00736)	0.0894*** (0.0227)	0.110*** (0.00764)	0.0980*** (0.0213)
Log unskilled	0.583*** (0.0179)	0.658*** (0.0427)		
Log low skilled			0.307*** (0.0197)	0.352*** (0.0332)
Log medium skilled			0.228*** (0.0116)	0.247*** (0.0287)
Log skilled	0.131*** (0.0105)	0.0676** (0.0293)	0.129*** (0.0120)	0.0844*** (0.0298)
Broadband × Intercept	−0.618*** (0.181)	−0.765 (0.550)	−0.835*** (0.173)	−0.961** (0.468)
Log capital	0.00774 (0.0111)	0.0212 (0.0312)	−0.00572 (0.0109)	0.0125 (0.0310)
Log unskilled	−0.0297 (0.0215)	−0.133** (0.0604)		
Log low skilled			−0.0340* (0.0185)	−0.100* (0.0512)
Log medium skilled			0.0396*** (0.0135)	0.0174 (0.0450)
Log skilled	0.0910*** (0.0111)	0.195*** (0.0435)	0.0851*** (0.00756)	0.160*** (0.0439)
Firm-year observations	16,744	16,744	16,250	16,250
	p-values			
Test for no skill bias	.000	.000	.000	.000

Notes. * $p < .10$, ** $p < .05$, *** $p < .01$. This table uses the survey sample of joint-stock firms over the period 2001–2007. The dependent variable is the log value added. OLS estimates are based on equation (3), whereas IV estimates are based on equations (3) and (4). Sampling weights are used to ensure representative results for the population of joint-stock firms. (Un)Skilled comprises workers with(out) a college degree. Low skilled comprises individuals without high school diploma and medium skilled consists of high school graduates (without a college degree). All regressions include fixed effect for year, municipality, and industry. The standard errors are clustered at the municipality level and robust to heteroskedasticity. The hypothesis tests report p -values from two-sided tests of the null hypothesis that the coefficient on broadband × log skilled is equal to the coefficient on broadband × log unskilled (or availability × log low skilled).

5.8 Table VII: IV evidence that adoption changes the production function

Read: This is the core structural result.

- In the two-skill IV specification, broadband adoption changes the output elasticities by about −0.133 for unskilled labor and +0.195 for skilled labor.
- In the three-skill IV specification, broadband adoption changes the output elasticities by about −0.100 for low-skilled labor, +0.017 for medium-skilled labor, and +0.160 for skilled labor.
- The interaction with capital is small and insignificant.

Interpretation. This is strong evidence of skill-biased technological change, not just scale expansion.

Caution. The IV estimates are larger than the OLS ones, but the paper warns that the difference can reflect both endogeneity bias and heterogeneity because the IV is local to early adopters.

TABLE VIII
WAGE REGRESSIONS WITH INTERACTIONS BETWEEN TASKS AND BROADBAND AVAILABILITY

Dependent variable	(1)	(2)	(3)
	Log hourly wage		
	Skill categories		
		2 skill levels	3 skill levels
Abstract	0.371*** (0.0142)	0.283*** (0.0139)	0.272*** (0.0140)
Routine	-0.0641*** (0.00653)	-0.0664*** (0.00573)	-0.0700*** (0.00577)
Manual	0.0248*** (0.00791)	0.0156** (0.00769)	0.0138* (0.00740)
Availability $\times$ Abstract	0.173*** (0.0320)	0.157*** (0.0298)	0.157*** (0.0297)
Availability $\times$ Routine	-0.0357*** (0.00798)	-0.0344*** (0.00766)	-0.0338*** (0.00791)
Availability $\times$ Manual	0.00200 (0.0115)	0.00145 (0.0107)	0.00273 (0.0104)
Worker-year observations	4,586,333	4,586,333	4,586,333
Controlling for educational attainment:			
Skill levels	No	Yes	Yes
Availability $\times$ Skill levels	No	Yes	Yes
Tests for no task bias:		<i>p</i> -values	
Equality of abstract and routine	.000	.000	.000
Equality of abstract and manual	.000	.000	.000
Equality of manual and routine	.041	.040	.036

Notes. * $p < .10$, ** $p < .05$, *** $p < .01$. We consider workers aged 18-67 over the period 2001-2007 who are recorded in the wage statistics survey and for which we observe occupation codes at the four-digit level. The occupation codes are linked with measures of task intensity from the Dictionary of Occupational Title (DOT), as reported by Autor and Dorn (2013). Following Autor, Levy, and Murnane (2003), we convert the DOT measures into percentiles of the task distribution. Column (1) presents results from a regression of worker-year observations of log hourly wages on task intensities and their interaction with broadband availability in the local labor market. Column (2) adds indicator variables for two levels of skill and their interaction with broadband availability. Column (3) includes indicator variables for three levels of skills and their interaction with broadband availability. (Un)Skilled comprises workers with(out) a college degree. Low skilled comprises individuals without high school diploma and medium skilled consists of high school graduates (without a college degree). All regressions include fixed effects for year, municipality, and industry and controls for gender, years of experience, and years of experience squared. The standard errors are clustered at the municipality level and robust to heteroskedasticity. We report *p*-values from two-sided tests of the null hypothesis that the coefficients on the interaction variables of the different task intensities are equal.

5.9 Table VIII: broadband complements abstract tasks and substitutes for routine tasks

Read: The key coefficients are on the interactions between availability and job tasks.

- Availability $\times$ Abstract is about 0.157 to 0.173 and highly significant.
- Availability $\times$ Routine is about -0.034 to -0.036 and highly significant.
- Availability $\times$ Manual is near zero.

Translate the magnitude. A 10 percentage point increase in availability raises wages for workers with abstract-task intensity at the 75th percentile by about 0.9% relative to workers at the 25th percentile, and lowers wages for routine-task-intensive jobs by about 0.2% on the same comparison.

Why it matters. This is the most direct evidence on *how* broadband changes production. It does not merely reward schooling; it rewards the kinds of tasks that ICT complements.

6 Short summary

- The paper uses a staggered public broadband rollout in Norway to identify the labor-market and productivity effects of broadband internet.
- In reduced form, broadband availability raises skilled wages and employment, lowers low-skilled

wages, and raises the output elasticity of skilled labor while lowering that of low-skilled labor.

- In IV production-function estimates, broadband adoption appears to be a skill-biased technological change: it increases the marginal importance of skilled labor and reduces that of low-skilled labor.
- The first stage is strong: more local household availability causes more firm adoption.
- The exclusion restriction is supported by placebo tests and by evidence against alternative explanations involving product-demand shifts, telecom or IT firms, generalized PC adoption, and work-from-home channels.
- The task-based interpretation is central: broadband complements nonroutine abstract tasks, substitutes for routine tasks, and leaves manual tasks largely unaffected.
- The paper's broader message is that ICT infrastructure can raise productivity while widening skill differences inside the labor market.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that broadband internet is not a neutral productivity shock. In the Norwegian setting, it looks much more like a skill-biased technological change. The public rollout increased broadband availability, availability induced firm adoption, and adoption shifted the production technology in a way that raised the productivity and labor-market outcomes of skilled workers while hurting low-skilled workers.

7.2 Economic intuition

Broadband helps more with tasks involving information processing, communication, coordination, and problem solving than with routine work. Firms that adopt it can reorganize production around those tasks. Skilled workers, who are already concentrated in abstract-task occupations, become more productive and therefore more valuable. Low-skilled workers, who are disproportionately in routine jobs, become less valuable when broadband allows firms to automate, standardize, or reorganize those tasks.

The paper also suggests that firms capture a nontrivial share of the surplus. Wages move in the same direction as marginal productivities, but much less one-for-one in the short run.

7.3 Takeaway

One sentence to remember this paper, it is this: **broadband internet raised productivity, but it did so by complementing skilled and abstract-task labor while substituting away from routine, lower-skill labor.**

That is why the paper matters for both productivity policy and distributional policy. Infrastructure that looks universally growth-enhancing on average may still create winners and losers inside the labor market.